

Real GDP Growth Nowcasting:

Advanced Toolkit for
Bahamian Public Sector
Decision-Makers



COURSE OVERVIEW

This intensive training program is designed for public sector professionals involved in national economic policy and statistics. It aims to equip participants with theoretical foundations and practical skills to utilize an advanced nowcasting toolkit in **Python** and **R**. The course leverages real-time structured and unstructured data sources to generate timely estimates of economic activity, thereby enhancing informed policymaking and resource allocation. It is built upon the methodology outlined in our working paper “*Nowcasting Real GDP Growth in The Bahamas*”, which develops a data-driven nowcasting framework for small, open economies facing data availability challenges.

Target audience

❖ **The Bahamas National Statistical Institute (BNSI):**

Economists, statisticians, and data analysts involved in data collection, compilation, and economic indicator production.

❖ **Ministry of Finance and Central Bank:**

Economists, fiscal and monetary analysts, and policy advisors responsible for macroeconomic analysis, budget planning, and economic forecasting.

Prerequisites

- ❖ Basic economic and statistics concepts and indicators.
- ❖ Familiarity with quantitative analysis.
- ❖ Laptop with Python and R installed.
 - [See here](#) for **python**
 - [See here](#) for **R**
- ❖ Prior experience with data manipulation (e.g., in Excel, Stata, EViews, or basic programming) is beneficial but not strictly required.

AGENDA

<i>Time</i>	Tuesday <i>Nov 25</i>	Wednesday <i>Nov 26</i>	Thursday <i>Nov 27</i>
🕒 9:00 – 10:45am	Session 1: Python fundamentals	Session 3: Estimation and variable selection	Session 5: The Nowcasting tool
🕒 10:45 – 11:00am	<i>Break</i>	<i>Break</i>	<i>Break</i>
🕒 11:00 – 12:45pm	Session 1: R Fundamentals	Session 3: Estimation and variable selection	Session 5: The Nowcasting tool output
🕒 12:45 – 1:15pm	<i>Lunch break</i>	<i>Lunch break</i>	<i>Lunch break</i>
🕒 1:15 – 3:00pm	Session 2: Website scraping	Session 4: Nowcasting Theory	Session 5: The Nowcasting tool output
🕒 3:00 – 3:15pm	<i>Break</i>	<i>Break</i>	<i>Break</i>
🕒 3:15 – 5:00pm	Session 2: Scraping satellite and google data	Session 4: Nowcasting Theory	

INSTALLING PYTHON



CONTENT OUTLINE

Session 1: Python and R Fundamentals for Economic Analysis

🕒 **Duration:** 4 hours

❖ **Mode:** Face-to-Face

❖ **Learning Objectives:**

- Familiarize with the basic concepts of Python and R
- Identify data structures
- Run Python/R script in each language
- Perform basic data uploading
- Perform basic data cleaning

❖ **Content:**

- *Introduction to Python for Economics:*
 - Setting up the *jupyter* environment
 - Installing packages in python Anaconda
 - Data structures: integers, float, string, list, dictionary, and DataFrames
 - Conditionals
 - Looping: for and while loops
 - Basic functions
 - Importing/exporting various data formats (txt, csv, xlsx)
 - Basic data selection, filtering, and aggregation.
 - Pandas data types: dates and numerics
 - Pandas data reshape: merge, melt, pivot
- *Introduction to R for Economic Data:*
 - RStudio interface and basic R syntax.
 - Data structures (data.frame).
 - Introduction to tidyverse for data manipulation (dplyr).

❖ **Activities:**

- Loading a small dataset (e.g., GDP and stock index) into Python and R.
- Performing common data cleaning tasks (e.g., standardizing date formats, merging, reshape).
- Calculating basic descriptive statistics.
- Time series plot

Session 2: Scraping economic data and data cleaning

🕒 **Duration:** 4 hours

❖ **Mode:** Face-to-Face

❖ **Learning Objectives:**

- Retrieve traditional and non-traditional data from web sources
- Apply basic download and analysis to satellite imagery and Google search trends.
- Perform data cleaning using standard economic adjustments: deflation, seasonal adjustment, and growth rates.
- Estimate a simple Ordinary Least Square model to macroeconomic data

❖ **Content:**

- *Scraping data*
 - Accessing macroeconomic indicators from the web and API
 - Accessing press release archives and textual data
 - Satellite Nighttime Lights (NTL): principles, data sources (e.g., NOAA DMSP/VIIRS). Downloading and using satellite imagery data in Python
 - Google Trends: Using Python script for search activity related to consumption, employment, business sentiment. Downloading and loading Google trends data
- *Preprocessing economic data*
 - Discuss data publication lags, frequency, and revisions.
 - Deflation of Nominal Values using CPI:
 - Seasonal Adjustment
 - Period-Over-Period Growth Rate
 - Running OLS in python

❖ **Activities:**

- Source and retrieve sample data (e.g., nightlights for a specific area using NASA's website, Google Trends for key economic terms, and a small table from a public sector website).
- Data cleaning exercise on a dataset.

Session 3: Variable Selection for Economic Models

🕒 **Duration:** 4 hours

❖ **Mode:** Face-to-Face

❖ **Learning Objectives:**

- Apply Least Angle Regression (LARS) for data-driven variable pre-selection in a high-dimensional context.
- Use Principal Component Analysis (PCA) for dimensionality reduction.
- Apply the nowcasting environment (ragged edges, missing value) to a data series
- Estimate a simple nowcasting with OLS model

❖ **Content:**

- *Variable pre-selection with LARS*
 - Theory and practical application: selecting variables by incrementally building a linear model
 - Evaluating LARS performance against a baseline OLS
- *Dimensionality Reduction with PCA*
- *The Nowcasting tool's pipeline:*
 - Missing value imputation
 - Lagged data and publication lag adjustment
 - Data flattening for non-time series models
 - Inclusion of autoregressive lags

❖ **Activity:**

- Participants will prepare a dataset for a simple Nowcasting (missing value imputation, simulate data availability, and perform data flattening)
- Participants will run an estimator (OLS) in the dataset.

Session 4: Nowcasting Models: Theory and Application for Economic Estimation

🕒 **Duration:** 4 hours

❖ **Mode:** Face-to-Face

❖ **Learning Objectives:**

- Understand the theoretical basis of prominent nowcasting estimators, encompassing both traditional econometric and advanced machine learning techniques.
- Recognize the strengths and weaknesses of different models for specific economic nowcasting challenges.
- Gain conceptual understanding of how these models are implemented within the nowcasting tool, including handling real-time data limitations.

❖ **Content:**

- *Key Nowcasting Model Families for Economic Activity:*
 - *Econometric Models:*
 - Autoregressive Moving Average (ARMA)
 - Ordinary Least Squares (OLS)
 - Regularization Models (LASSO, Ridge, Elastic Net)
 - Vector Autoregression (VAR)
 - Mixed-Data Sampling (MIDAS) regressions
 - Machine Learning MIDAS
 - Dynamic Factor Models (DFM)
 - *Machine Learning Models:*
 - Decision Tree (DT) regression
 - Random Forest (RF)
 - Gradient Boosting Tree (GBT)
 - Long Short-Term Memory (LSTM) networks
- *Model implementation in Python:*
 - walkthrough of model-specific functions, parameters and options in the nowcasting tool's interface

Session 5: Nowcasting Tool execution and results interpretation

🕒 **Duration:** 4 hours

❖ **Mode:** Face-to-Face

❖ **Learning Objectives:**

- Navigate and run the core components of the nowcasting tool, starting with a fundamental OLS example
- Understand how the tool's architecture scales from simple models to a complex ensemble
- Evaluate and use the nowcast output

❖ **Content:**

- *Main Nowcasting code architecture and workflow: a detailed tour of the tool's core script*
- *Data input formats, configuration parameters, and output directories.*
- *Practical set up of a simple example with OLS: step-by-step guided execution*
 - Data uploading and pre-processing
 - Train-test split and evaluation strategy for a single model
 - Model training and in-sample evaluation for the OLS estimate
 - out-of-sample point forecast generation
- *Scaling Up: running the code with various specifications and estimators*
 - Transition from building block to an ensemble
 - Discuss the implications of running this intensive process: strengths and limitations
 - Demonstrating how the tool manages and executes this large-scale, automated nowcasting process.
- *Assessing nowcast output: accuracy and forecast*
 - Retrieving predictions
 - Visualization and tables

❖ **Activities:**

- Participants will execute the full nowcasting code for one specification, excluding LSTM (all other models included)
- Participants will visualize results
- Q&A and open discussion on tool integration.

READING LIST & RESOURCES

Python

Installation: <https://www.youtube.com/watch?v=mg6cMkz9Q0c>

Brief introduction (10mins): <https://www.youtube.com/watch?v=fWjsdhR3z3c>

Complete introduction (1hr): <https://www.youtube.com/watch?v=kqtD5dpn9C8>

Data cleaning: https://www.youtube.com/watch?v=bDhvCp3_IYw

Web scraping: <https://www.youtube.com/watch?v=8dTpNajxaH0>

Exploratory Data Analysis: <https://www.youtube.com/watch?v=Liv6eeb1VfE>

A full course (11hr): <https://www.youtube.com/watch?v=wUSDVGivd-8>

R

Installation: <https://www.youtube.com/watch?v=mxNrU902uyc>

Brief Introduction: <https://www.youtube.com/watch?v=FY8BISK5DpM>

Complete introduction: <https://www.youtube.com/watch?v=BvKETZ6kr9Q>

Exploratory Data Analysis: https://www.youtube.com/watch?v=3iz-2iM4RFE&list=PLtL57Fdbwb_ChN-dNR0qBjH3esKS2MXY3&index=11

Data cleaning: <https://www.youtube.com/watch?v=sV5lwAJ7vnQ>

Data wrangling: <https://www.youtube.com/watch?v=kg9Kgvv9-0g>

Full course (for beginners): https://www.youtube.com/playlist?list=PLtL57Fdbwb_ChN-dNR0qBjH3esKS2MXY3

Nighttime Lights

Addison, D. M., & Stewart, B. (2015). Nighttime lights revisited: the use of nighttime lights data as a proxy for economic variables. *World Bank Policy Research Working Paper*, (7496).

Gibson, J., Olivia, S., Boe-Gibson, G., & Li, C. (2021). Which night lights data should we use in economics, and where?. *Journal of Development Economics*, 149, 102602.

World Bank Light Every Night (Guide) <https://worldbank.github.io/OpenNightLights/wb-light-every-night-readme.html>

Google Trends

Choi, H., & Varian, H. (2012). Predicting the present with Google Trends. *Economic record*, 88,2-9.

Cebrián, E., & Domenech, J. (2024). Addressing google trends inconsistencies. *Technological Forecasting and Social Change*, 202, 123318.

Varian, H. (2023). Nowcasting with google trends. *Economic Record*, 99, 1-12.

Estimation and Variable Selection in Python

Chinn, M. D., Meunier, B., & Stumpner, S. (2023). *Nowcasting world trade with machine learning: a three-step approach* (No. w31419). National Bureau of Economic Research.

Efron, B., Hastie, T., Johnstone, I., & Tibshirani, R. (2004). Least angle regression.

Hopp, D. (2024). Benchmarking econometric and machine learning methodologies in nowcasting GDP. *Empirical Economics*, 66(5):2191–2247.

Sargent, T. J., & Stachurski, J. (2019). Lectures in Quantitative Economics with Python. [See here.](#)

Nowcasting Models

ARMA:

QuantEcon: Forecasting an AR(1) Process in Python
https://python.quantecon.org/ar1_turningpts.html#

Linear models:

QuantEcon: Linear Regression in Python <https://python.quantecon.org/ols.html>

Python: https://scikit-learn.org/stable/modules/linear_model.html

OLS:

StatQuest: <https://www.youtube.com/playlist?list=PLblh5JKOoLUlzaEkCLIUxQFjPIlapw8nU>

Python library: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html

Ridge regression:

StatQuest: <https://www.youtube.com/watch?v=Q81RR3yKn30> Python library: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.Ridge.html

Hoerl, A. E. and Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67.

LASSO:

StatQuest: <https://www.youtube.com/watch?v=NGf0voTMlcs>

Python library: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.Lasso.html

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B (Methodological)*, 58(1):267–288.

Elastic net:

StatQuest: <https://www.youtube.com/watch?v=1dKRdX9bfIo>

Python library: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.ElasticNet.html

Zou, H. and Hastie, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 67(2):301– 320.

Mixed-data sampling (MIDAS):

R library: <https://cran.r-project.org/web/packages/midasr/midasr.pdf>

Ghysels, E., Santa-Clara, P., and Valkanov, R. (2004). The midas touch: Mixed data sampling regression models. *Cirano working papers*, CIRANO.

Ghysels, E., Kvedaras, V., and Zemlys, V. (2016). Mixed frequency data sampling regression models: The r package midasr. *Journal of Statistical Software*, 72(4).

Mixed Frequency Bayesian Vector Autoregression (MF-BVAR):

R library: http://r.meteo.uni.wroc.pl/web/packages/mfbvar/vignettes/mfbvar_jss.pdf

Ankargren, S. and Jonéus, P. (2020). Simulation smoothing for nowcasting with large mixedfrequency vars. *Econometrics and Statistics*.

Schorfheide, F. and Song, D. (2015). Real-time forecasting with a mixed-frequency var. *Journal of Business Economic Statistics*, 33(3):366–380.

Cimadomo, J., Giannone, D., Lenza, M., Monti, F., and Sokol, A. (2020). Nowcasting with large bayesian vector autoregressions. *ECB Working Paper Series 2453*, European Central Bank.

Dynamic Factor Model (DFM):

Bok, B., Caratelli, D., Giannone, D., Sbordone, A. M., and Tambalotti, A. (2018). Macroeconomic nowcasting and forecasting with big data. *Annual Review of Economics*, 10:615–643.

Hopp, D. (2024). Benchmarking econometric and machine learning methodologies in nowcasting GDP. *Empirical Economics*, 66(5):2191–2247.

Giannone, D., Reichlin, L., and Sala, L. (2005). Monetary policy in real time. In Gertler, M. and Rogoff, K., editors, *NBER Macroeconomics Annual 2004*, volume 19, pages 161–200. The MIT Press, Cambridge, MA.

Stock, J. H. and Watson, M. W. (2002b). Macroeconomic forecasting using diffusion indexes. *Journal of Business & Economic Statistics*, 20(2):147–162.

Decision Tree Regression:

StatQuest: <https://www.youtube.com/watch?v=g9c66TUylZ4&t=904s>

Python library: <https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeRegressor.html>

Breiman, L., Friedman, J., Stone, C. J., and Olshen, R. A. (1984). *Classification and Regression Trees*. CRC press.

Random Forest Regression:

StatQuest: https://www.youtube.com/watch?v=J4Wdy0Wc_xQ&t=143s Python library: <https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html>

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.

Gradient Boosted Tree Regression:

StatQuest: <https://www.youtube.com/watch?v=3CC4N4z3GJc> Python library: https://scikit-learn.org/stable/auto_examples/ensemble/plot_gradient_boosting_regression.html

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5):1189–1232.

Long-Short Term Memory (LSTM):

StatQuest: <https://www.youtube.com/watch?v=YCzL96nL7j0>

LSTM tutorial: <https://www.youtube.com/watch?v=c0k-YLQGKjY>

Hopp, D. (2021). Economic Nowcasting with Long Short-Term Memory Artificial Neural Networks (LSTM). UNCTAD Research Paper No. 62, United Nations Conference on Trade and Development.

Hochreiter, S. and Schmidhuber, J. (1997). Long short-term memory. Neural computation, 9(8):1735–1780.

MIDAS-ML:

R library: <https://cran.r-project.org/web/packages/midasml/midasml.pdf>

Babii, A., Ghysels, E., and Striaukas, J. (2022). Machine learning time series regressions with an application to nowcasting. Journal of Business Economic Statistics.

FACILITATORS



José Luis Saboin is an Economics Specialist in the Caribbean Country Department of the Inter-American Development Bank (IDB), where he holds the position of Country Economist for The Bahamas. Previously, José Luis worked in the Andean Country Department of the IDB and before that in the Western Hemisphere Department of the International Monetary Fund (IMF). He has led and co-led multiple research and data collection projects across Latin America and The Caribbean. His areas of research and interest include development macroeconomics, international monetary economics, fiscal policy in resource-rich countries, nowcasting and technology adoption, which have been published at the IMF, IDB and other peer-reviewed journals. He has an Economics degree from the Andrés Bello Catholic University in Venezuela, a

Master in Economic Policy Management from Columbia University in New York City (USA) and a PhD in Economics from George Mason University in Virginia (USA).



Diego A. Guerrero is a Lead Economist at Northwestern University's School of Education and Social Policy. He is an applied microeconomist specializing in public and labor economics. His recent work focuses on data-driven nowcasting estimators for economic forecasting, federal in-kind grants to police departments, and disability policy in the United States. He has also studied the effects of social programs on labor market outcomes, intergenerational mobility, and human capital formation in the Americas. His research has been featured in *Economics of Education Review*, a forthcoming book chapter in *The Legacy of James M. Buchanan*, and a revise-

and-resubmit at *Rehabilitation Counseling Bulletin*. Diego holds a PhD and an MA in Economics from the University of Tennessee, Knoxville, an MSc in Applied Economics

from St. Cloud State University, and a BA in Liberal Studies from Universidad Metropolitana. He also serves as a consultant to state and local governments and the Inter-American Development Bank, helping policymakers apply empirical evidence and state-of-the-art technologies in data science and forecasting.



Angelo Mazzocca is an Economics Consultant at The Caribbean Department of the Inter-American Development Bank. He holds a BS in Economics from Universidad Católica Andrés Bello (UCAB) in Venezuela, where he graduated with honors as second in his class. He specializes in development economics and Fintech. His professional and entrepreneurial efforts are centered on applying technology and economic analysis to foster growth and development at the country and firm levels, including his work on a Startup project for a Multi-service Recharge App in Caracas, Venezuela. ‘

